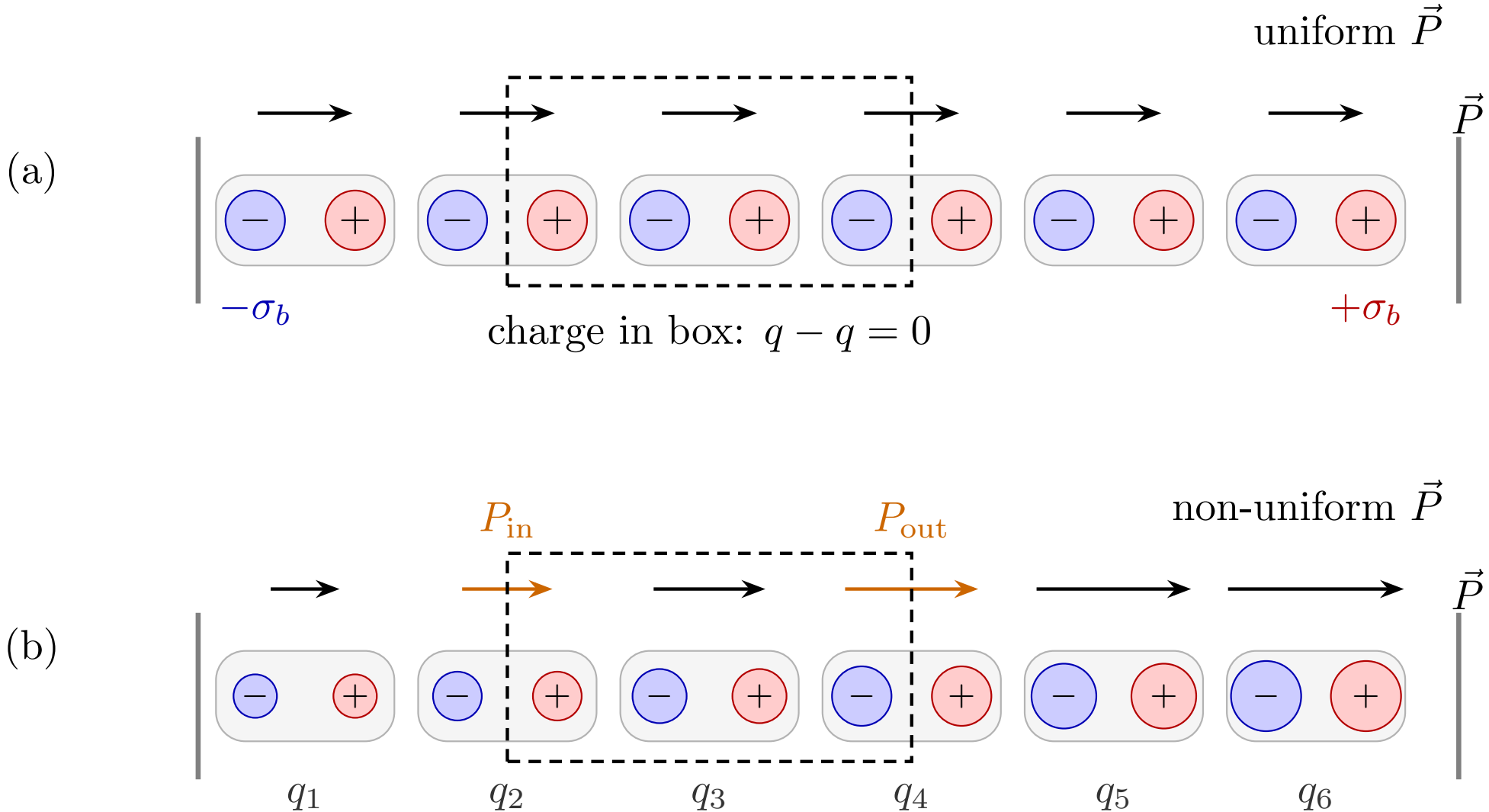

uniform $\vec{P}$
(a)
$\vec{P}$
$-\sigma_b$
$+\sigma_b$
charge in box: $q - q = 0$
non-uniform $\vec{P}$
P_{in}
P_{out}
(b)
$\vec{P}$
q_1
q_2
q_3
q_4
q_5
q_6
charge in box: $q_2 - q_4 = -(P_{\text{out}} - P_{\text{in}})\, A = -(dP/dx)\, \Delta V < 0$